

THE UNIVERSITY OF CHICAGO

**‘STRAIT TO CATASTROPHE’: THE ROLE OF PERCEPTUAL
FACTORS IN US-CHINA-TAIWAN DEFENSE POSTURE AND
NUCLEAR ESCALATION**

UCHICAGO EXISTENTIAL RISK LABORATORY
SUMMER FELLOWSHIP CAPSTONE PROJECT

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CHICAGO, ILLINOIS
AUGUST 2023

Introduction

Taiwan is the most indomitable problem in Sino-American relations and the most plausible theater of conflict between the two great powers. The “Taiwan problem” has existed for decades but is today more volatile and potentially damaging than ever. Rising tensions across the Taiwan Strait, paired with the People’s Republic of China’s rapidly increasing military capabilities, point to an uncertain future of the island and the balance of power between the United States (US) and China.¹ Further, common understandings of the Taiwan problem are rapidly losing credibility in the face of a changed world wherein China’s conventional and nuclear capabilities are on track to rival those of the US. This article combines an analysis of China’s military capabilities in a Taiwan scenario with robust examination of perceptual factors in Sino-American relations that may complicate escalation management. This research aims to understand the role that perceptual factors play in China’s defense posture and risk of nuclear escalation over Taiwan.

When evaluating China’s nuclear posture, it is critical to note the perceptual roles of not just China but also the US. In nuclear and security studies, especially analyses produced by US analysts reliant on American data and references, there exists a prevailing idea that the US is a fundamental protector of the global order, and any military or nuclear development from countries the US considers strategic opponents is viewed as, at minimum, threatening. Beyond that, there is often an assumption that international security crises will develop primarily because of logical and rational choices. But as Kristensen et al. note, the US operates large numbers of silo-based missiles and systems designed to protect its retaliatory capabilities and insists this posture is necessary and stabilizing.² It is entirely reasonable, then, that China may seek a similar posture. Yet Chinese nuclear development is met by American skepticism and concern, contributing to conversations about the “pacing challenge” of China, or a new Cold War framework for the US-China relationship. This contributes to the foreign policy concept of mutual constitution or reciprocity, which Kyle Haynes and Brandon Yoder explain is the “degree to which one state’s behavior shifts to match another’s previous actions.”³ In the US-China case, this means that when the US treats China as something threatening which must be contained, China in turn will become something

¹ Throughout this paper I will refer to the People’s Republic of China simply as China. The Republic of China will be referred to as Taiwan.

² Kristensen et al., “Chinese Nuclear Weapons 2024,” 182.

³ Haynes, Kyle, and Brandon K. Yoder. “Trust, Cooperation, and the Tradeoffs of Reciprocity.” *Conflict Management & Peace Science* 41, no. 1 (January 1, 2024): 26–46.

the US must contain. Analyses which sensationalize Chinese military and nuclear development beyond practical applications tend to do more harm than good by contributing to this damaging framework. However, it is still valuable to analyze China's nuclear modernization and rapidly rising conventional military capabilities, especially in the context of the Taiwan problem.

This article proceeds in four sections, beginning with relevant background information and a review of the literature on perceptual factors in international relations and, more specifically, the US-China context. The second section draws on open-source data to analyze China's changing military capabilities, focusing on China's nuclear development. The third section introduces perceptual factors as a frame of analysis, opening with an exploration of the potential breakdown of deterrence and operationalizing three main facets of perception: signaling, security dilemma and entanglement, and wartime perceptual shift. The final section frames the implications of conventional war and nuclear war between the US and China, as well as of Chinese control of Taiwan, then concludes with recommendations to inform future policies and research.

In general, this analysis finds that perceptual factors play a key role in Sino-American relations, especially over Taiwan. Moreover, analysis of potential outcomes identifies catastrophic risk that are driven by these perceptual factors, and that exist at an intensity such that mitigation must be prioritized. This article highlights multiple pathways to tandem escalation and de-escalation, indicating that the Taiwan issue is highly complex and should be carefully considered through broad analytical lenses and multiple theoretical frameworks.

Background

Before turning to an overview of nuclear deterrence and theories of perception and political psychology, it is important to lay out some relevant historical context regarding the cross-strait situation.

Taiwan is strategically important to the US and China for a multitude of reasons spanning military strategy, nationalism, and global power competition. In order to understand China's perception and defense posture towards Taiwan, it is crucial to first understand the importance that both the US and China place upon the island. Many articles examine Taiwan's economic significance deriving from its position as the global leader in semiconductor production and its political significance as a democratic "beacon" near rising Communist China. My work does not

aim to restate this—instead, I rely on previous work which outlines Taiwan’s importance to the global order and, uniquely, to China, and highlight that Taiwan is the most likely theater of conflict between the US and China.

China’s goal of reunification with Taiwan today is grounded in history, particularly the Qing dynasty, and every Chinese leader since Mao Zedong has proclaimed China’s rightful sovereignty over the island (Yan & Mao 2016). Reunification as a goal has shaped China’s strategic posture and policies for decades. But Taiwan, self-governed since 1949, remains politically and economically distinct from China. Since his appointment as the General Secretary of the Chinese Communist Party in 2012 and subsequent rise to President of China in 2013, Xi Jinping too has maintained reunification with Taiwan as one of China’s key future goals. Importantly, Chinese leaders are very clear on Taiwan’s importance. It is not speculative to argue that Taiwan is important to China, nor that China would go to great lengths to reunify with Taiwan. Taiwan is narratively important to China such that China’s growing military and nuclear capabilities are directly applicable to analyzing the potential of a conflict between the US and China over Taiwan.

The US relationship with Taiwan is less rhetorically clear than that of China. Instead, it is grounded in a policy known as strategic ambiguity, which does not recognize Taiwan’s sovereignty and independence diplomatically. It also maintains the US ability to choose to defend Taiwan without committing to a military intervention in the event of a Chinese attack on Taiwan. This is underpinned by the Taiwan Relations Act of 1979, in which the US commits to provide Taiwan with defense capabilities while not guaranteeing military intervention. The US is Taiwan’s main international supporter, and Taiwan is still reliant on the US for military defense. This means that the US maintains a sort of dual deterrence wherein the threat of US military support of Taiwan prevents China from launching an invasion, and the threat of losing US support prevents Taiwan from endangering the status quo, for example by declaring independence.⁴ According to Nien-Chung Chang-Liao and Chi Fang, strategic ambiguity has historically allowed the US to “maintain cooperative and beneficial relations with both sides of the Taiwan Strait, contributing to peace and prosperity in the region.”⁵ In short, American foreign policy toward Taiwan is designed to prevent Chinese control of Taiwan while minimizing the risk of destabilizing the status quo.

⁴ Kuo, Raymond. “Strategic Ambiguity Has the U.S. and Taiwan Trapped.” *Foreign Policy* (January 2023).

⁵ Chang-Liao, Nien-Chung, and Chi Fang. “The Case for Maintaining Strategic Ambiguity in the Taiwan Strait.” *Washington Quarterly* 44, no. 2 (June 1, 2021): 45. doi:10.1080/0163660X.2021.1932088.

The US and China have distinct and opposing claims regarding the future of Taiwan. These claims, paired with mounting tension across the Taiwan Strait precipitated Chinese military action towards Taiwan in 1954, 1958, and 1996, marking the three major Taiwan Strait crises. None of the crises saw direct military combat operations between the US and China, but in each crisis, the US backed Taiwan militarily and played a major role in deescalating the conflict.⁶ These crises underscore the persistent volatility in cross-strait relations and the geopolitical importance of Taiwan.

Cross-strait diplomacy is no stranger to imbalance, but analyses indicate tensions are rising and military and political actions, too, are increasing accordingly. The 2023 China Military Power Report (CMPR), issued by the US Department of Defense, lists increasing Chinese military coercion as one of the key takeaways from China's course of national strategy. Emphasizing the "more dangerous, coercive, and provocative" actions China undertakes in the Indo-Pacific region, the CMPR notes over 180 instances of "coercive and risky air intercepts against U.S. aircraft in the region" as well as 100 instances of the same behavior against US allies.⁷ This verbiage is not used in the 2020 CMPR, indicating coercive air actions are increasingly notable in 2023. This increase may be motivated by changes in US policy, including Nancy Pelosi's visit to Taiwan or the US drift from the One China Policy, but a marked increase is nonetheless notable. The CMPR also highlights increased Chinese pressure against Taiwan. "Throughout 2022, [China] increased provocative and destabilizing actions in and around the Taiwan Strait, to include ballistic missile overflights of Taiwan, significantly increased flights into Taiwan's self-declared air defense identification zone, and conducted a series of large-scale military exercises around Taiwan."⁸

Military provocation and coercion from China are contextualized within an increasingly authoritative political regime, Since Xi Jinping rose to power in 2012, China has become internationally domineering and domestically repressive.⁹ It tacitly challenges the current institutional international order the US has embraced since World War II. Xi has also maintained markedly stronger rhetoric surrounding reunification with Taiwan than previous Chinese leaders.

⁶ Chen, Yu-Hua. "China's Buffer Thinking towards Taiwan." *International Journal of Taiwan Studies* 6, no. 2 (July 1, 2023): 345–67. <https://doi.org/10.1163/24688800-20221268>.

⁷ US Department of Defense, "Annual Report 2023," x.

⁸ US Department of Defense, "Annual Report 2023," 136.

⁹ Garwin, Richard L., and Frank N. von Hippel. "How to Avoid Nuclear War with China." *Bulletin of the Atomic Scientists*, 79, no. 2 (March 2023): 57. doi:10.1080/00963402.2023.2178161.

In 1973, Chairman Mao told Secretary Kissinger that “[China] can do without Taiwan...let it come after one hundred years.”¹⁰ But in 2013 and 2019, Xi stated that the issue of Taiwan cannot be passed down generationally, applying more urgency and imminence to the issue than his predecessors.¹¹ Since Xi’s appointment, the rhetoric he and other Chinese officials use surrounding nuclear weapons too is aggressive and direct, embracing topics of potential war and violence. In 2017, researchers from Xi’s Rocket Force wrote that China should seek “nuclear forces sufficient to...ensure that our country can win the initiative in future wars.”¹² As tensions continue to rise, Taiwan’s future is perilous.

Taiwan’s strategic significance and ongoing turbulence set it as the most important location for a potential US-China conflict to occur, and rising cross-strait tensions indicate the “Taiwan problem” is more important than ever.

Literature Review

Although many sets of international relations theory are inherently rationalist and assume an effective implementation of strategically logical policy in pursuit of clearly understood goals, the real world is understood to be more complicated.¹³ Perceptual factors play a crucial role in shaping political decisions, particularly in the realm of international relations and security studies. Scholars have long debated the extent to which perceptions, misperceptions, and cognitive biases influence the behavior of states and policymakers. Two key figures in this debate are Robert Jervis and Christopher Fetscher, who have contributed significantly to the understanding of perceptual factors in international politics.

Robert Jervis is one of the most influential scholars in the study of perception in international relations. His monumental 1976 book, *Perception and Misperception in International Politics*, lays the foundation for understanding how cognitive processes and psychological factors influence decision-making at the state level. Jervis argues that the way states perceive each other's intentions, capabilities, and actions is often distorted by cognitive biases, leading to misperceptions

¹⁰ “Memorandum of Conversation between Mao Zedong and Henry Kissinger.” November 12, 1973. *Wilson Center Digital Archive*.

¹¹ Sacks, David. “What Xi Jinping’s Major Speech Means For Taiwan.” *Council on Foreign Relations* (July 2021).

¹² Buckley, Chris. “Fear and Ambition Propel Xi’s Nuclear Acceleration.” *New York Times* (2024).

¹³ This realist sentiment is echoed across myriad authors, notably: John Mearsheimer, *The Tragedy of Great Power Politics*. (New York: W. W. Norton & Company, 2001).

that can escalate conflicts or hinder cooperation. He explains that perceptions and reality do not always align, and it is “impossible to explain crucial decisions and policies without reference to the decision-makers’ beliefs about the world and their images of others.”¹⁴ Jervis identifies several key perceptual biases such as the “attribution error,” where states are likely to attribute hostile intentions to their adversaries’ actions while seeing their own actions as defensive.¹⁵ Another is the “mirror-imaging” bias, where states project their own values and beliefs onto others, assuming other states will behave similarly under similar circumstances.¹⁶ Jervis’ work emphasizes that perceptual factors are not errors but are often deeply rooted in the psychological makeup of decision-makers.

Alongside Jervis, Christopher Fettweis is a key scholar who applies perceptual factors to military strategy and foreign policy within the framework of a state. In his 2015 article “Misreading the Enemy,” Fettweis argues that states often create an image of the enemy from negative assumptions, leading them to misinterpret the intentions and capabilities of their adversaries.¹⁷ Often, this is founded on the assumption that differences between states are “fundamental and existential” and cannot be resolved.¹⁸ Perception is distorted further within a state when organizational and doctrinal factors within military and policy institutions that prioritize worst-case scenarios and aggressive postures.¹⁹ By outlining multiple flawed assumptions states repeatedly make in evaluating foreign states, Fettweis argues that perceptual errors are not purely individual but are often institutionalized within the military and state apparatus, leading to systemic miscalculations in foreign policy.

Both Jervis and Fettweis emphasize the importance of perception in shaping international relations. Of course, there are numerous authors who apply perceptual factors to international relations and even US-China relations specifically. Many of the authors I rely on in this paper, including Caitlin Talmadge and Christopher Twomey, highlight the importance of perceptual

¹⁴ Robert Jervis, *Perception and Misperception in International Politics*. (Princeton, New Jersey: Princeton University Press, 1976), 28.

¹⁵ Robert Jervis, *Perception and Misperception*, 321.

¹⁶ Robert Jervis, *Perception and Misperception*, 323.

¹⁷ Fettweis, Christopher. “Misreading the Enemy.” *Survival*, 57, no. 5 (2015): 149.

¹⁸ Fettweis, “Misreading the Enemy,” 151.

¹⁹ Fettweis, “Misreading the Enemy,” 163.

factors in Sino-American relations.²⁰ My paper utilizes these impactful works as foundational to understanding perceptual factors in international relations, yet extends their application to consider the changing landscape of the China-Taiwan-US relationship.

A New Order: The Shifting US-China Power Balance

Since the US and China nearly came to blows over Taiwan in 1996, the balance of power between the two nations has dangerously shifted.²¹ Every previous US-China confrontation has occurred in a world where the US maintains a dominant military, global influence, and economic system. But today's world sees a China that has emerged as a rapidly growing global power seeking, at minimum, military and status parity with the US, and, at maximum, a refiguring of the global system to support a new hegemony of Chinese power. Thus, we can no longer truly understand the US-China relationship through mere extrapolation from historic events, trends, and choices. Today more than ever, understanding the balance of power between the US and China requires modern analyses of military capabilities and changing strategy.

A wealth of recent literature catalogues China's rise in military power relative to the US, marking a worryingly indefinite upward trajectory of capabilities. The US-China Military Scorecard, a report compiled by RAND in 2015, examines China's modernization by comparing the US and China's military capabilities stretching from 1996 to 2017 and assessing the relative advantage or disadvantage of each nation in varying conflict scenarios.²² In a "Taiwan Scenario" which occurs geographically near China, Heginbotham et. al lay out nine potential operational areas of conflict.²³ In 1996, the US enjoyed "major advantages" in six out of the nine, an advantage in one, and approximate parity in another. China had only one advantage out of nine.²⁴ By 2017,

²⁰ Perceptual factors are frequently mentioned in: Talmadge, Caitlin. "Would China Go Nuclear? Assessing the Risk of Chinese Nuclear Escalation in a Conventional War with the United States." *International Security*, 41, no. 4 (2017): 50-92. Doctrinal differences, state-based decision-making, and perceptual distortions in Sino-American relations are thoroughly analyzed in: Christopher Twomey, *The Military Lens: Doctrinal Difference and Deterrence Failure in Sino-American Relations* (Cornell University Press, 2011).

²¹ Diamond, Larry, and James O. Ellis Jr. "Deterring a Chinese Military Attack on Taiwan." *Bulletin of the Atomic Scientists* 79, no. 2 (March 2023): 65-71. <https://doi.org/10.1080/00963402.2023.2178166>.

²² Heginbotham, Eric, Michael Nixon, Forrest E. Morgan, Jacob L. Heim, Jeff Hagen, Sheng Tao Li, Jeffrey Engstrom, Martin C. Libicki, Paul DeLuca, David A. Shlapak, David R. Frelinger, Burgess Laird, Kyle Brady, and Lyle J. Morris. "The U.S.-China Military Scorecard: Forces, Geography, and the Evolving Balance of Power, 1996-2017." *RAND Corporation* (2015): xxii.

²³ Heginbotham et al., "The US-China Military Scorecard," xxix.

²⁴ Ibid.

the US maintained only three of nine advantages, with no major advantages.²⁵ Approximate parity existed in four areas, and China enjoyed the advantage in two.²⁶ This data stops in 2017, but it is likely that a similar assessment today may see China's advantage in up to eight areas of conflict, and the US advantage in no more than two. Heginbotham's recent data-rich wargaming alongside Matthew Cancian and Mark Cancian finds that defending Taiwan in the event of a Chinese invasion comes at a tremendous military cost that the US has not faced since World War II.²⁷ The US military advantage over China in Taiwan is rapidly eroding, and the modern balance of power between the two nations is shifting at an unprecedented rate. As the outcome of a conventional war becomes increasingly uncertain, the relative likelihood of conflict and potential nuclear escalation trends dangerously upward.

One of the most worrisome elements of the changing power balance between the US and China is China's nuclear development. The 2020 China Military Power Report (CMPR) estimated China's nuclear stockpile around 200 warheads and projected a doubling of capabilities over the next decade, reaching 400 warheads by 2030.²⁸ Just three years later, the 2023 CMPR estimated China's nuclear stockpile at over 500 operational nuclear warheads.²⁹ The same report projected possession of over 1,000 operational nuclear warheads by 2030, more than doubling the 2020 CMPR's estimate for the same year.³⁰ The 2022 CMPR makes the same prediction for 2030, extending further to project China's stockpile growth to 1,500 warheads by 2035.³¹ Although this figure is not repeated in the 2023 CMPR, the report does predict "[continued] growth".³² China's nuclear development significantly exceeds US projections of China's nuclear capacity and forces based on previous data from only a few years ago, indicating that a shift in the balance of power between the two nations is not just imminent but ongoing. The US currently retains nuclear superiority as a result of its amassed nuclear stockpile estimated at 3,708 nuclear warheads,

²⁵ Ibid.

²⁶ Ibid.

²⁷ Cancian, Mark, Matthew Cancian, and Eric Heginbotham. "The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan." *Center for Strategic & International Studies International Security Program* (January 2023): 87.

²⁸ US Department of Defense. "Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2020." *Office of the Secretary of Defense* (September 2020): ix.

²⁹ US Department of Defense. "Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2023." *Office of the Secretary of Defense* (October 2023): viii.

³⁰ Ibid.

³¹ US Department of Defense. "Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2022." *Office of the Secretary of Defense* (November 2022): 94.

³² US Department of Defense, "Annual Report 2023," viii.

although many of these are not operational on a regular basis.³³ But the growth of China's arsenal places future nuclear superiority in question. Based on its nuclear development and modernization, China's nuclear arsenal is likely the fastest-growing in the world.³⁴

A key aspect of China's nuclear development is the prolonged modernization of its land-based nuclear-capable missile force and sub-surface maritime force. This is evident in the construction of 350 new missile silos across the deserts of northern China and the mountains of central-eastern China.³⁵ Paired with the ongoing construction of numerous new bases for land-based mobile missile launchers, these new developments add to China's existing 350 launchers for land-based nuclear-capable missiles and heavily extend their short-, medium-, and intermediate-range missile launch capabilities.³⁶

Alongside its rapidly growing land-based nuclear missile capabilities, China's tandem modernization of submarines will likely improve the nation's nuclear survivability and attack capability by great lengths. Historically, China's two generations of nuclear ballistic missile submarines, Type 092 and Type 094, were "noisy" because of their design and propulsion systems (including their large missile storage "humps") and thus relatively easy to detect with existing US Sound Surveillance System (SOSUS) undersea acoustic detection infrastructure.³⁷ The Type 094, while quieter than the Type 092, still produces more noise than Cold War era US and Soviet submarines. But China is actively working to quiet their submarines. The improved 094A submarine was publicly revealed in 2018, and the third-generation Type 096 is under development.³⁸ In 2023, China launched a nuclear-powered attack submarine with its propeller replaced by a pump-jet propulsion system, effectively reducing a portion of submarine noise.³⁹ As China continues to fund modernization efforts and apply the same technology used on American submarines to its own, Bin Li and Riqiang Wu write that "the level of submarine silence will

³³ Hans M. Kristensen, Matt Korda, Eliana Johns, and Mackenzie Knight. "United States Nuclear Weapons, 2024." *Bulletin of the Atomic Scientists*, 80, no. 3 (2024): 182, <https://doi.org/10.1080/00963402.2024.2339170>.

³⁴ Hans M. Kristensen, Matt Korda, Eliana Johns, and Mackenzie Knight. "Chinese Nuclear Weapons, 2024." *Bulletin of the Atomic Scientists*, 80, no. 1 (2024): 51, <https://doi.org/10.1080/00963402.2023.2295206>.

³⁵ Kristensen et al., "Chinese Nuclear Weapons," 58.

³⁶ Ibid.

³⁷ Li, Bin, and Riqiang Wu. "U.S. Strategy of Damage Limitation vis-à-vis China: Long-term Programs and Effects." *China International Strategy Review*, 6 (2024): 15.

³⁸ Li & Wu, "U.S. Strategy of Damage Limitation," 15.

³⁹ Gale, Alastair. "The Era of Total U.S. Submarine Dominance Over China is Ending." *The Wall Street Journal* (November 20, 2023).

inevitably improve over time.”⁴⁰ China is one of only four known nations to maintain a “nuclear triad” consisting of the capabilities to launch nuclear weapons from land-, air-, and sea-based delivery systems.⁴²

China’s large-scale modernization of military technology does not happen in a political vacuum; instead, it occurs in tandem with China’s dangerous slide toward risky and potentially threatening behavior which endangers the maintenance international peace especially over the Taiwan Strait. China’s increasing provocation and signaling towards Taiwan is indicative of the importance of Taiwan to China. While it is likely that a key driver for China’s nuclear force development is increasing its survivability and seeking a degree of qualitative and quantitative parity with US and Russian nuclear forces, China’s increasing action towards Taiwan clearly emphasizes the role of a potential conflict over Taiwan as a key reason to improve China’s nuclear capabilities.⁴³

China’s increase in military capabilities and provocative action is accompanied by heightening tensions in mainland China and across the Taiwan Strait, as discussed in a previous section. As China continues to modernize its military forces and capabilities, the long-standing perceived US military dominance over China is in question. The future is uncertain, but one thing is clear: China is more ready and able today to fight and potentially win, or at least do considerable damage, in a war with the United States over Taiwan. Basing predictions for Taiwan on historic Taiwan Strait crises or previous US-China relations is quickly losing merit, and the outcome of a Taiwan conflict is dangerously uncertain.

⁴⁰ Li & Wu, “U.S. Strategy of Damage Limitation,” 15.

⁴¹ However, speculation that China is developing submarine technology that renders US surveillance mechanisms obsolete is rooted in sensationalism and a limited scope of undersea surveillance networks. SOSUS, developed during the Cold War to surveil Soviet submarines, uses passive acoustic technology to detect submarine-generated narrowband tonals. As Soviet submarines improved and became quieter over time, the US implemented the Fixed Distribution System (FDS) in key ocean chokepoints, using upward-facing sensors to detect even extremely quiet submarines that pass overhead. To build a robust undersea surveillance network against China, the US also developed the Transformational Reliable Acoustic Path System (TRAPS) which acts similarly to FDS and monitors passing submarines. Even if Chinese submarines advance rapidly and become extremely quiet, FDS and TRAPS infrastructure will still be able to detect Chinese submarines that pass through key areas of surveillance. See Li & Wu, “U.S. Strategy of Damage Limitation,” 15-17.

⁴² Asano, Hideo. “China’s Nuclear Expansion and Implications for U.S. and Global Security.” *Nuclear Threat Institute Atomic Pulse* (July 2023).

⁴³ David Logan and Phillip Saunders. *Discerning the Drivers of China’s Nuclear Force Development: Models, Indicators, and Data* (Washington, D.C., National Defense University Press, 2023), 2.

Perceptual Factors

Mutual Deterrence: Grounds for Confidence and Fear

This section lays out the concept of mutual deterrence as it stands between the US and China, then draws upon China's recent nuclear expansion to cast the long-standing assumption of deterrence into question. Finally, I will explain why the potential costs of a breakdown of mutual deterrence are severe to the extent that, regardless of how unlikely the actual occurrence is, it must be carefully investigated.

In general, the likelihood of conventional war between the US and China over Taiwan is relatively low. After all, the Taiwan problem has existed for nearly 75 years, but US-China tensions, while increasing, have yet to escalate to war. And even in the event of a conventional war between the two nations, Chinese nuclear escalation, too, is unlikely. This is because of the long-standing status quo of *mutual deterrence*. Essentially, the US maintains a minimum threshold of conventional capabilities needed to deter China from taking military action because the threat from the US is severe. Theoretically, China does the same, deterring the US from military action by maintaining a credible threat should a conflict occur. This is duplicated in the nuclear sphere, and nuclear policy globally is dominated by deterrence theory, in which each nuclear nation theoretically disincentivizes the others from nuclear use. In basic terms, if only one nation had nuclear weapons, they may become globally dominant because no other nation can pose a real threat to its military superiority. But if many nations have nuclear weapons, no one nation is incentivized to use its nuclear capabilities because there is a sure and sudden retaliation from other nuclear nations. Mutual deterrence is thus relatively reliable to prevent a large-scale war between China and the US since it is highly likely that both nations would suffer massive losses even in a purely conventional war.⁴⁴

China also practices a strategy termed "Active Defense" or, informally, "counter-deterrence," which aims to do exactly what the name signals: prevent the US from successfully employing deterrence to prevent China from pursuing its legitimate goals and interests.⁴⁵ This approach employs diverse methods including cyber warfare and space capabilities to weaken US power projection and alliances to in turn prevent the US from deploying its full power in the event

⁴⁴ Cancian, Cancian, and Heginbotham, "The First Battle of the Next War," 87.

⁴⁵ Radzinsky, Brian. "The Strategic Implications of the Evolving US-China Nuclear Balance." *The Washington Quarterly*, 44, no. 4 (2022): 165

of a conflict.⁴⁶ China's practice of counter-deterrence, while operating very near to the principles of basic deterrence, importantly signifies that this form of deterrence is specifically engineered to counter the US, instead of general deterrence which works against any strategic competitor.

Another dimension of US-China relations that further decreases the relative likelihood of war is China's long-standing no-first-use policy. Since China's emergence as the world's fifth nuclear-armed state in 1964, it has repeatedly reaffirmed this commitment to never be the *first* to escalate to nuclear use in a conflict. Currently, of the nine nuclear nations, only China and India formally maintain this doctrine.⁴⁷ This means that relative to a conflict involving Russia or Pakistan, for example, a conflict involving China is less likely to reach nuclear escalation.⁴⁸ However, though China has not officially altered its long-standing policy, recent significant shifts in its strategic posture and nuclear arsenal (as outlined in the previous section) raise questions about the validity of this policy. Analysts, scholars, and policymakers alike debate how the US should make sense of China's nuclear expansion, arriving at a fundamental disagreement of whether these changes threaten nuclear deterrence.⁴⁹ When the viability of mutual deterrence is a matter of debate, the chance of a failure of deterrence, however small, is nonzero. This should be worrying.

Assuming that a framework like mutual deterrence is perfect and will indisputably prevent a large-scale conflict between the US and China would be a mistake. Scholars who adopt this view tend to ignore the human factor inherent in both peacetime and wartime decision making, and falsely assume that political decisions will be made by perfectly rational actors acting within the presumed framework of deterrence. Even in "perfect conditions" where deterrence is unquestionably reliable, there remains a (potentially infinitesimal) chance of a failure of deterrence. In the world of today, where China is undergoing significant nuclear expansion and tensions over the Taiwan Strait are increasing, this small chance is extremely concerning. The dangers posed by nuclear escalation and a breakdown of mutual deterrence, regardless of the cause, are potentially

⁴⁶ Ibid.

⁴⁷ Havrén, Sari Arho. "China's No First Use of Nuclear Weapons Policy: Change or False Alarm?" *The Royal United Services Institute for Defence and Security Studies* (October 2023).

⁴⁸ Rittenhouse, Brendan, and Caitlin Talmadge. "Then What? Assessing the Military Implications of Chinese Control of Taiwan." *International Security*, 47, no. 1 (2022): 7.

⁴⁹ Radzinsky, "Strategic Implications," 163.

catastrophic. Thus, the potential paths to a US-China war and nuclear escalation over Taiwan must be carefully evaluated.

Signaling & Chinese Reactions

To China, and in a lesser extent to the US, Taiwan is important militarily, economically, and narratively. It is not unfounded to argue that Taiwan is the single most likely cause for a large-scale US-China conflict.⁵⁰ Traditional understandings of cross-strait relations examine Taiwan's military and political significance and its implications on US and Chinese grand strategy, yet often miss the critical role perceptual factors play in the cross-strait balance. Studying perceptual factors in international relations involves studying how cognitive psychology applies to political decision making, stretching back to Robert Jervis' historic work *Perception and Misperception in International Politics*. [DEFINE PERCPETUAL FACTORS HERE] In this analysis I will operationalize perceptual factors in three main sectors: signaling, entanglement and survivability, and wartime perception shifts. Each of these three facets of perception are multidimensional and complex. Thus, I will rely on a mix of tangible examples and theoretical propositions to draw out the importance of perceptual factors in cross-strait tensions and US-China relations as a whole.

One of the most tangible perceptual factors in Taiwan relations today is political and military signaling. Signaling, at its purest form, is *how states communicate desired information and images of themselves and their intentions*.⁵¹ Signals by their very nature are meant to be seen, heard, or perceived by their recipient as the signaler intends, but in reality this does not always occur. There are almost limitless pathways for signaling to fail—i.e., to be misinterpreted or miscommunicated—such as incorrect decoding or interpretation, lack of credibility, faulty communication, negative actor motivation, and discounting.⁵² In short, for something intended to increase communication, signals are incredibly easy to misinterpret. But moving towards a better understanding of signaling is not impossible, and one of the key requirements to improve signal interpretation is acknowledging previous failures and accepting that not all signals can or should

⁵⁰ Rittenhouse & Talmadge, "Then What?" 7.

⁵¹ Balestrieri, Brendan. "Disrupting Deterrence Signaling: Examining the Fifth Wave of Technology's Impact." *Journal of Strategic Security*, 16, no. 2 (2023): 1-10.

⁵² Balestrieri, "Disrupting Deterrence Signaling," 1.

be taken at face value. But the goal of this paper is not to prescribe a better way to interpret signals; instead, I will analyze a key example of the collision of American and Chinese signaling and perception—Nancy Pelosi’s visit to Taiwan—to highlight the centrality of Taiwan to China’s narrative and the increasing danger posed by US threats to this narrative.

This example is well-known and heavily discussed in news media and US-China-Taiwan relations articles, carrying implications for not just the US-China balance but for the Indo-Pacific region as a whole: US House of Representatives Speaker Nancy Pelosi’s visit to Taipei on August 2, 2022.⁵³ In the immediate aftermath of Pelosi’s trip, both China and the US argued that the *other* nation was responsible for the crisis, with China claiming that the US provoked it and the US arguing that China’s response was an overreaction. Chinese foreign minister Wang Yi stated, “...the basic facts are that the United States first provoked China on the Taiwan question and blatantly violated China’s sovereignty and territorial integrity.”⁵⁴ The US in turn argued that China’s unprecedented militaristic response, including short-range ballistic missile overflights of Taiwan and repeated overflights of the median line separating Taiwanese and Chinese territory, was a “manufactured” crisis with the goal of coercing not just Taiwan but also the international community.⁵⁵ As the nations fought for control of the narrative, tensions in the Taiwan Strait peaked at a level not seen in nearly thirty years.⁵⁶

Regardless of which nation is ultimately responsible, this event, sometimes referred to as a “fourth Taiwan Strait crisis” exemplifies how fiercely fundamental the issue of Taiwan is to China. Pelosi’s visit to Taiwan, regardless of whether the US intended it as a signal (in all likelihood it should be understood as an intentional signal), was interpreted as a signal by China and elicited a strong reaction consisting of retaliative signaling. China’s reaction to Pelosi’s visit to Taiwan is instructive, beginning with insight into how future similar scenarios are likely to play out. Another key takeaway from Pelosi’s visit is the importance of preamble, reaction, and

⁵³ Jude Blanchette, Charles Edel, Christopher B. Johnstone, Scott Kennedy, Victor Cha, Ellen Kim, and Gregory B. Poling. “Speaker Pelosi’s Taiwan Visit: Implications for the Indo-Pacific.” *Center for Strategic and International Studies* (August 2022).

⁵⁴ Primary source now deleted, text appears as cited in Blanchette et al., “Speaker Pelosi’s Taiwan Visit.”

⁵⁵ Seligman, Laura. “Pentagon Still Assesses China Will Not Invade Taiwan in Next 2 Years.” *Politico* (August 8, 2022).

⁵⁶ Haenle, Paul, and Nathaniel Sher. “How Pelosi’s Taiwan Visit Has Set a New Status Quo for U.S.-China Tensions.” *Carnegie Endowment for Peace* (August 17, 2022).

narrative-building in signaling. When major events occur and the battle for narrative control ensues, signaling easily becomes muddled, and some signals rise in prominence while others fall.

A key facet of military and political signaling which is often forgotten in case-based analyses is the hard-to-predict cognitive factor of perception. Robert Jervis writes that it is “impossible to explain crucial decisions and policies without reference to the decision-makers’ beliefs about the world and their images of others.”⁵⁷ Even when national context is well integrated into signaling perception, the individuals making or communicating decisions are constantly navigating the inherent differences between perception and reality.⁵⁸ Signaling is not uniformly perceived, and the exact reaction of a country or individual to any given signal is uncertain and cannot be guaranteed. Regardless of the signal the US intended to communicate by sending Speaker Pelosi to Taiwan, China interpreted it as a threatening, inflammatory, and coercive action. The same can be said for the US perception of China’s reactive military overflights of Taiwan. Nations react more strongly to what they perceive to be threatening signals than to rhetorical signals intended to officially maintain peace and stability in the region, that is, to uphold the status quo. China’s strong reaction to Pelosi’s visit indicates that future US encroachment on Chinese sovereignty of Taiwan will be met with an abrupt and severe reaction. On one hand, the fact that China did not further escalate to direct military conflict can be seen as an indication that China would not escalate to war over Taiwan. But on the other hand, China’s severe reaction indicates that even a signal as small as a US delegation to Taiwan will be taken as a threat to Chinese interests and lead to rising cross-strait tensions. China’s reaction to Pelosi’s visit to Taiwan should be worrying, and the US must carefully consider its future signals in light of the near crisis.

Security Dilemma & Entanglement

Running parallel to the concept of mutual deterrence is the *security dilemma*, a framework of international relations wherein one state’s security enhancement (such as increasing military strength) inadvertently leads to a sense of insecurity in the other state, which undergoes its own security enhancement, leading to a cycle of escalating tensions and hostilities.⁵⁹ In the security

⁵⁷ Robert Jervis, *Perception and Misperception in International Politics*. (Princeton, New Jersey: Princeton University Press, 1976), 28.

⁵⁸ Jervis, *Perception and Misperception*, 29.

⁵⁹ Tiong, Wei Jie, and Mingjiang Li. “The US-China Security Dilemma: The Need for Constant Mitigation.” *S. Rajaratnam School of International Studies, Nanyang Technical University* (November 2023): 1.

dilemma, both states have only defensive motivations. But because states are distrustful of other states' actions and intentions, the cyclical security dilemma continues. In certain cases, the intensity of the security dilemma increases; in the case of the US-China security dilemma, the geographic position of Taiwan as very near to China intensifies the security dilemma. Because the location of Taiwan would, in the event of a US-China conflict over Taiwan, place China at a massive strategic advantage, the US further develops its military and security strategies to maintain a credible threat against China over Taiwan, leading China to in turn increase its security to counter the growing capabilities of the US. It is important to note that the security dilemma here is not tripolar between the US, China, and Taiwan, it is bipolar between only the US and China.⁶⁰ Taiwan is caught in the middle, serving not as an equivalent participant in a military-driven security dilemma but instead as a potential theater wherein a US-China clash may occur.

The US-China security dilemma is unique compared to each country's other international relations because American and Chinese strategic posture are heavily influenced by each other. For example, China's nuclear strategy over the past few decades has been primarily motivated by ensuring secure second-strike capability to counter its assessments of US nuclear capabilities.⁶¹ Inversely, since 2017 the US has "identified China as the top pacing challenge" driving US strategic posture and development.⁶² When two nations spiral into direct competition, the potential consequences of the security dilemma worsen. Especially because the US and China are two of the three modern "great powers," rising tensions over the past 5 years have cast the two nations into what some have termed a second Cold War, extending past military and strategic policies into a trade war, economic competition, and great power rivalry.⁶³ Both Joe Biden and Xi Jinping overtly reject the conception of US-China competition as a Cold War, but the troubling similarities between modern US-China relations and Cold War-era US-USSR relations mean that regardless of designating an official "second Cold War," it is likely that US-China competition will define global relations in the foreseeable future.⁶⁴

⁶⁰ Kuo, Raymond, Hunzeker, Michael, and Mark Christopher. "Scared Strait: How Should American Deter China from Attacking Taiwan?" *Foreign Affairs* (February 2024).

⁶¹ Hiim, Henrik, M. Taylor Fravel, and Magnus Langset Trøan. "The Dynamics of an Entangled Security Dilemma: China's Changing Nuclear Posture." *International Security*, 47, no. 4 (Spring 2023): 149.

⁶² Ratner, Ely. Quoted in: Clark, Joseph. "U.S. Focuses on Deterrence as China Raises Stakes in Indo-Pacific." *U.S. Department of Defense News* (October 2023).

⁶³ Huang, Yukon. "The U.S.-China Trade War Has Become a Cold War." *Carnegie Endowment for International Peace* (September 2021).

⁶⁴ Alperovitch, Dmitri. "How the U.S. Can Win the New Cold War." *Time Magazine* (May 1, 2024).

The already concerning posture of the US-China security dilemma is elevated further by the alarming interaction of nuclear and conventional capabilities, especially in China. The US and China are embroiled in an “entangled security dilemma” wherein nuclear and conventional capabilities are increasingly intertwined, leading to altered threat perceptions.⁶⁵ This is dangerous because when nuclear and conventional threats and capabilities are intermingled, negative shifts in the conventional balance can lead to a state increasingly relying on nuclear weapons as compensation for conventional vulnerability. In turn, the opposing state will likely view this increased prioritization of nuclear weapons as threatening to its own nuclear capabilities and increase nuclear development to combat this threat.⁶⁶ Like the conventional security dilemma, this leads to a downward spiral of constantly increasing threats and development, but worse than the conventional security dilemma, elevation to the nuclear realm carries a much higher potential for catastrophe.

In an entangled security dilemma, the fear that a conventional assault has the potential to weaken or destroy nuclear retaliatory capabilities may alter a state’s perception of vulnerability, creating yet another pathway to cyclical bipolar nuclear development. Alternately, this may lead to the state’s development of conventional capabilities to increase the survivability of its nuclear forces.⁶⁷ Both nuclear and conventional development in this case lead to an increased threat perception for the other state.

The framework of an entangled security dilemma is primarily relevant to US-China competition because great power competition between two of the nine nuclear states uniquely provides a theater for competition, whether perceived or direct, to play out in real time. One of the largest potential threats which serves as a trigger for cyclical nuclear and conventional development is China’s significant overlap of nuclear and conventional capabilities. Chinese submarines, space assets, and missiles have dual nuclear and conventional capabilities, and both conventional and nuclear-armed assets are stored together.⁶⁸ Perhaps more dangerous is China’s dual use of centralized command and control systems. Thomas Christensen writes, “if strikes by

⁶⁵ Hiim, Fravel, and Trøan, “The Dynamics of an Entangled Security Dilemma,” 149.

⁶⁶ Ibid.

⁶⁷ Ibid.

⁶⁸ Christensen, Thomas J. “The Meaning of the Nuclear Evolution: China’s Strategic Modernization and US-China Security Relations.” *Journal of Strategic Studies*, 35, no. 4 (August 2012): 470. doi:10.1080/01402390.2012.714710.

the United States on China's conventional coercive capabilities or their critical command and control nodes and supporting infrastructure were to appear in Beijing as a conventional attack on its nuclear retaliatory capability or as a precursor to a nuclear first strike, even a China that generally adheres to a No-First-Use posture might escalate to the nuclear level.”⁶⁹ China's uncertain second-strike capability means that in the event of a conventional war with the US, the risk of China's nuclear escalation is significantly heightened.

Insecurity & Paranoia: Second-Strike Security & Wartime Perception

China's physical overlap of nuclear and conventional capabilities, paired with yet unchanged US nuclear superiority, paves the way for Chinese concern that its second-strike capabilities are not secure. In the event of a conventional conflict over Taiwan, Joshua Rovner writes: “The targets in the hypothetical first strike [from the US] would include China's ballistic missiles and fixed and mobile launchers, as well as...facilities for targeting and guidance.”⁷⁰ Because China's DF-21 missiles are mounted with both conventional and nuclear warheads, US strikes against missiles and missile launchers would be extremely threatening to China.⁷¹ The same logic applies to command and control systems used by both conventional and nuclear operations.⁷² Rovner continues, “While U.S. planners might be confident that they can distinguish conventional from nuclear targets, Chinese officials might not be.”⁷³ This highlights the key role of perception in conflict: regardless of whether the US *actually* threatens China's second-strike capabilities with a conventional assault, if China *fears* that its retaliatory capabilities may be threatened, it could lead to Chinese nuclear escalation.

The potential escalatory threat posed by a conventional assault against Chinese forces is not new, and China has sought to secure its second-strike capabilities with nuclear development and modernization. Much of China's recent nuclear growth, as discussed in the previous section,

⁶⁹ Ibid.

⁷⁰ Rovner, Joshua. “AirSea Battle and Escalation Risks.” *Changing Military Dynamics in East Asia Policy Brief 12* (2012): 4.

⁷¹ Talmadge, “Would China Go Nuclear?,” 55.

⁷² Twomey, Christopher. “Asia's Complex Strategic Environment: Nuclear Multipolarity and Other Dangers.” *Asia Policy* (January 2011): 64.

⁷³ Ibid.

has promoted the security of its second-strike potential. Hans Kristensen and his coauthors explain that the construction of three new silo fields containing hundreds of new silos is a “significant shift in China’s nuclear posture...[caused by] a combination of strategic and operational objectives including protecting the retaliatory capability against a first strike.”⁷⁴ China is acutely aware of the weaknesses of its second-strike abilities, and it is highly likely that its recent nuclear development is motivated at least in part by seeking retaliatory security.

But perception, again, plays a key role in evaluating the efficacy of the strategies China has employed in this direction; China’s retaliatory capabilities are certainly elevated relative to prior years, but a secure second-strike is not yet guaranteed. A conventional assault by the US on Chinese dual use missile silos or command and control systems might endanger China’s retaliatory capabilities. China may then be triggered to escalate to nuclear use in a “use it or lose it” mentality for the remainder of its nuclear capabilities. Alternately, the fear of a conventional attack wiping out nuclear capabilities could lead China to a preemptive nuclear first use, assuming the threat is significant enough to warrant the reputational hit China’s international image would sustain by disobeying its no-first-use doctrine. In addition to the “use it or lose it” pressure potential, wartime decisions are inherently dangerous because of the rapidly accelerated timeline on which they must be made. During a conflict with the US, China’s ability to accurately assess US military actions is heavily inhibited, in part because the US will intentionally maintain ambiguity to damage China’s situational awareness.⁷⁵ This raises the risk potential in multiple areas, the most concerning of which is nuclear escalation. If China is unable to rapidly discern whether US weapons are conventional- or nuclear-armed, it may lead to China’s nuclear “retaliation” to an incorrectly interpreted conventional strike by the US. In short, there are myriad pathways to nuclear escalation. One of the most worrying elements of this framing is that instead of nuclear escalation remaining a far-off risk largely mitigated by mutual deterrence, there are direct channels to nuclear escalation.

Again, it matters less what is *objectively* real, and more what China *perceives* is real. In her 2017 “Will China Go Nuclear?” piece, Caitlin Talmadge encapsulates the conundrum: “The contest will come down to whether China believes it can hide a couple of dozen mobile missiles from the United States—or, more precisely, whether China believes that the United States believes

⁷⁴ Kristensen et al., “Chinese Nuclear Weapons 2024,” 63.

⁷⁵ Talmadge, “Would China Go Nuclear?,” 88.

that China can hide these missiles.”⁷⁶ This is why the perceptual aspect is so pertinent to understanding China’s nuclear risk. Perhaps China will be able to develop its nuclear retaliatory capabilities to ensure survivability against a first strike, or perhaps it will not. Either way, this downward spiral towards nuclear development, escalation, and potential first use is concerning. If China does not believe its second-strike potential is guaranteed, the “use it or lose it” problem may trigger preemptive use. The conversation is changing from conventional to nuclear war, and it is worrying. The point here is that mutual deterrence, while relatively reliable, is not perfect. There are logical and perceptual pathways to a failure of deterrence. In the event of a conventional fight over Taiwan between the US and China, nuclear escalation is unlikely but not impossible. That there is a nonzero chance of nuclear escalation, however small it may be, should be concerning.

The increasing chance of nuclear escalation in the event of a conventional war is exacerbated further by wartime perception and paranoia. This is because the relative possibility of nuclear escalation is heavily reliant on China’s *confidence*, both in the deterrence framework and in its nuclear capabilities. It is hard to be confident after suffering a first strike. Talmadge explains that it is one thing to be confident in mutual deterrence when the prospect of a damaging conventional assault is remote, but another thing entirely when a conventional assault has already occurred and threatened retaliatory capabilities.⁷⁷ Nuclear escalation rises in possibility when confidence falters. Wartime perceptual dynamics increase fear and decrease confidence, and according to Talmadge “could lead Chinese leaders to fear the worst...even if the war had not produced real changes in the U.S. ability to destroy China’s arsenal.” Even in peacetime, perception plays a key role in military and political decisions and has the potential to cause powerful negative shifts. In wartime, this power is magnified tenfold—fears and tensions run high, and threats are perceived as much more realistic after the initial deterrence has been breached.

The assumption that China, in the event of a conventional conflict, would be paranoid and risk nuclear escalation is based on China’s own management of its only nuclear crisis. In 1969, China’s border war with the Soviet Union escalated to the threat of a Soviet attack on China’s nuclear program. Talmadge explains that on three separate occasions, Chinese leaders were “sure that a Soviet nuclear attack was imminent.”⁷⁸ This attack never happened, but China prepared to

⁷⁶ Talmadge, “Would China Go Nuclear?,” 87.

⁷⁷ Talmadge, “Would China Go Nuclear?,” 88.

⁷⁸ Talmadge, “Would China Go Nuclear?,” 89.

use its nuclear weapons despite the vast nuclear asymmetry between itself and the Soviet Union and expectation that a nuclear war would lead to devastation for China.⁷⁹

Luckily, in this case, China's paranoia led to de-escalation. The Sino-Soviet border conflict never escalated to nuclear use. But China today is a radically different nation with a radically different nuclear capacity. The China of 1969, when faced with nuclear paranoia, de-escalated the conflict. But would today's China, equipped with a strong nuclear capacity and dominant conventional force rivaling that of the US, make the same choice? *Maybe*. Or China could choose to escalate, risking a global power war with nuclear use—in other words, catastrophe. It is unlikely that China would choose nuclear escalation, given the devastation it would face in a nuclear war. But it is not impossible. The question remains: are we willing to bet the future of the world on that chance, however small, never occurring? I argue it is reckless to assume nuclear war with China over Taiwan is impossible. In the context of an entangled security dilemma all but defined by perception and misperception, a conventional conflict between the US and China over Taiwan can quickly turn nuclear. This should be worrying.

Implications of War and Escalation

The pathways to a conflict between the US and China over Taiwan, and for that conflict to escalate to nuclear use, are clear and concerning. But to understand the imminence of mitigating such an event, we must accurately understand the implications. I will analyze the implications of three main situations: a conventional war between the US and China over Taiwan, a nuclear war in the same context, and Chinese control of Taiwan. I find that each of these scenarios are destabilizing to the global order. Unsurprisingly, nuclear war is the most devastating possibility, but each of the three situations are concerning such that policies of mitigation, stabilization, and de-escalation must be prioritized to maintain global security.

The first situation in my analysis is a conventional war between the US and China over Taiwan. The most likely path to this scenario is China commencing a military invasion to regain sovereignty over Taiwan (perhaps in response to some perceived provocation from an independence-minded leader in Taipei), followed by a swift US military response in defense of

⁷⁹ Talmadge, "Would China Go Nuclear?," 90.

Taiwan. As discussed earlier in this paper, Heginbotham and his coauthors find that China and the US have relative parity of military advantages in the Taiwan Strait.⁸⁰ This means that a conventional war between the two nations is unlikely to come to a rapid end, instead continuing over an extended period and causing massive losses of military forces for both China and the US. The detailed wargaming by Cancian, Cancian, and Heginbotham finds that a conventional war would come at a massive cost: “The United States and Japan lose dozens of ships, hundreds of aircraft, and thousands of servicemembers. Such losses would damage the U.S. global position for many years. While Taiwan’s military is unbroken, it is severely degraded and left to defend a damaged economy on an island without electricity and basic services. China also suffers heavily. Its navy is in shambles, the core of its amphibious forces is broken, and tens of thousands of soldiers are prisoners of war.”⁸¹ The authors find that the most likely result sees China’s invasion faltering and Taiwan remaining autonomous.⁸² However, the destabilization resulting from the conflict and the damaged US global position indicates the conflict would nonetheless be catastrophic. In addition, the significant damage to the global economy and halt to technological progress resulting from the likely destruction of Taiwan’s economy and semiconductor production facilities would create echoes of destabilization worldwide and long-term. As explained in the previous section, a conventional war also significantly raises the risk of nuclear escalation due to wartime perceptual factors and use-it-or-lose-it pressures, heightening the potential global impact by multiple orders of magnitude.

The second scenario of conflict is a nuclear war between the US and China precipitated by a conflict over Taiwan. This situation would likely occur following one of the pathways enumerated in the previous section, moving from a conventional war into a nuclear exchange. Assuming that China is the first to escalate to nuclear use, driven by misperception, paranoia, or secure second-strike preservation, the US is not likely to back down and would respond with nuclear retaliation.⁸³ The US retains nuclear superiority over China; while China’s nuclear capabilities are indisputably growing, in the near future there is very little chance that China would emerge as the victor of a nuclear exchange with the US. However, nuclear war has never occurred.

⁸⁰ Heginbotham et al., “The U.S.-China Military Scorecard,” xxiv.

⁸¹ Cancian et al., “The First Battle of the Next War,” 3.

⁸² Ibid.

⁸³ Talmadge, Caitlin. “The US-China Nuclear Relationship: Why Competition is Likely to Intensify.” *Brookings Global China Project* (September 2019): 2.

There is little understanding of which cities would be targeted by either the US or China, whether Taiwan would be caught in the crossfire and suffer nuclear destruction at a high level, and how many people would die. But one thing is certain: a nuclear war between the US and China would be catastrophic. Countless lives would be lost, cities destroyed, and the global order would be entirely destabilized. Potential resounding impacts including nuclear winter, dysfunctional communications systems, and collapsed trade routes would leave the world in bad shape even after the conclusion of nuclear exchange. In short, a nuclear war over Taiwan between the US and China would be catastrophic at a level the world has never seen. It must be avoided.

The final scenario I will analyze is Chinese control of Taiwan. This would likely result from one of two pathways: Chinese “peaceful” (peaceful here meaning non-conflict, though reunification could result from Chinese coercion) reunification with Taiwan, or a Chinese invasion of Taiwan where the US does not intervene on behalf of Taiwan. Reunification would result in political implications including damage to US alliances because of its loss of credibility after failing to defend Taiwan. In her 2023 article, “Then What? Assessing the Military Implications of Chinese Control of Taiwan,” Caitlin Talmadge examines the changing military balance resulting from China’s reunification with Taiwan. Talmadge finds that Chinese military control of Taiwan would lead to significant Chinese military advances including submarine warfare positioning that leaves US maritime forces much more vulnerable to attack, critical improvements in China’s “kill chain for long-range attacks against U.S. surface forces,” and strengthening China’s sea-based nuclear deterrent by leagues.⁸⁴ The potential cascading impacts of a perceived “weakened” US foreign policy include more Chinese expansion and coercive tactics, which may cast a dangerous nuclear shadow over future US-China conflicts. Thus, while Chinese control of Taiwan is vastly preferable to either of the prior scenarios, it would nonetheless be threatening to the stable preservation of the current global order.

Recommendations

Building an accurate and holistic understanding of China’s defense posture and likelihood of nuclear escalation over Taiwan necessitates inclusion of perceptual factors as a lens of analysis. By combining an analysis of China’s growing nuclear capabilities and perceptual factors in

⁸⁴ Talmadge, “Then What?” 10.

security relationships, I have outlined a key paradox: as China's nuclear forces grow, the risk of conflict is both increased and decreased, depending on which lens of analysis is adopted. In a straightforward analysis of military capabilities and a changing balance of power, China's increasing conventional and nuclear capabilities may empower China to take action to forcibly reunify with Taiwan, risking a conventional war and potential escalation to nuclear use. However, my analysis of second-strike survivability indicates that China's increasing nuclear capabilities may decrease the risk of a nuclear conflict by preventing China from preemptively escalating to nuclear use or retaliating to a conventional attack by the US with a nuclear strike. This paradox reveals that a one-to-one analysis of China's military capabilities and risk of conflict is reckless and does not account for perceptual factors. China's development of capabilities can be dangerous or safe, and disaggregating assumptions is critical to accurately understanding the impacts of changing military power. Policies and decisions regarding Taiwan *must* include consideration of perceptual factors.

In addition, my analysis has highlighted the necessity of broadening the scope of the US understanding of China's wartime perception and decision-making. There are clear pathways to conventional war, nuclear escalation, and a destabilized global order through the Taiwan Strait. The potential implications of any of these events are massive such that policies and decisions must consider risk mitigation as one of the key goals of Taiwan policy and Sino-American relations. China's military is changing, and the US must carefully consider the implications of these changes. China today is more capable than ever of waging and potentially winning a war with the US over Taiwan. This should be worrying. However, policies must be carefully evaluated to avoid negative spirals of reciprocity leading to increased destabilization. In short, the Taiwan problem is more concerning today than ever before, and the US must tailor its decisions and policies to preserve stability at all costs.

Conclusion

Perceptual factors play a key role in China's defense posture toward Taiwan and risk of nuclear escalation. China's military capabilities, both conventional and nuclear, are rapidly increasing and should be closely regarded by US policymakers. There are numerous pathways by which a conflict may occur between China and the US over Taiwan, and even more by which that conflict may turn nuclear. The implications of a conventional or nuclear conflict, or of Chinese

control of Taiwan, are dangerous and destabilizing such that US policy must be guided by risk mitigation. The future of the Taiwan Strait is uncertain. But that is not *necessarily* bad. It means that there is a possible path forward which avoids a crisis and improves stability in cross-strait relations and the global order. Taiwan may be the most likely theater for a US-China conflict to occur and spiral into catastrophic global power conflict. We must strive to understand the pathways to conflict and nuclear escalation in order to progress toward relative stability in the Taiwan Strait.